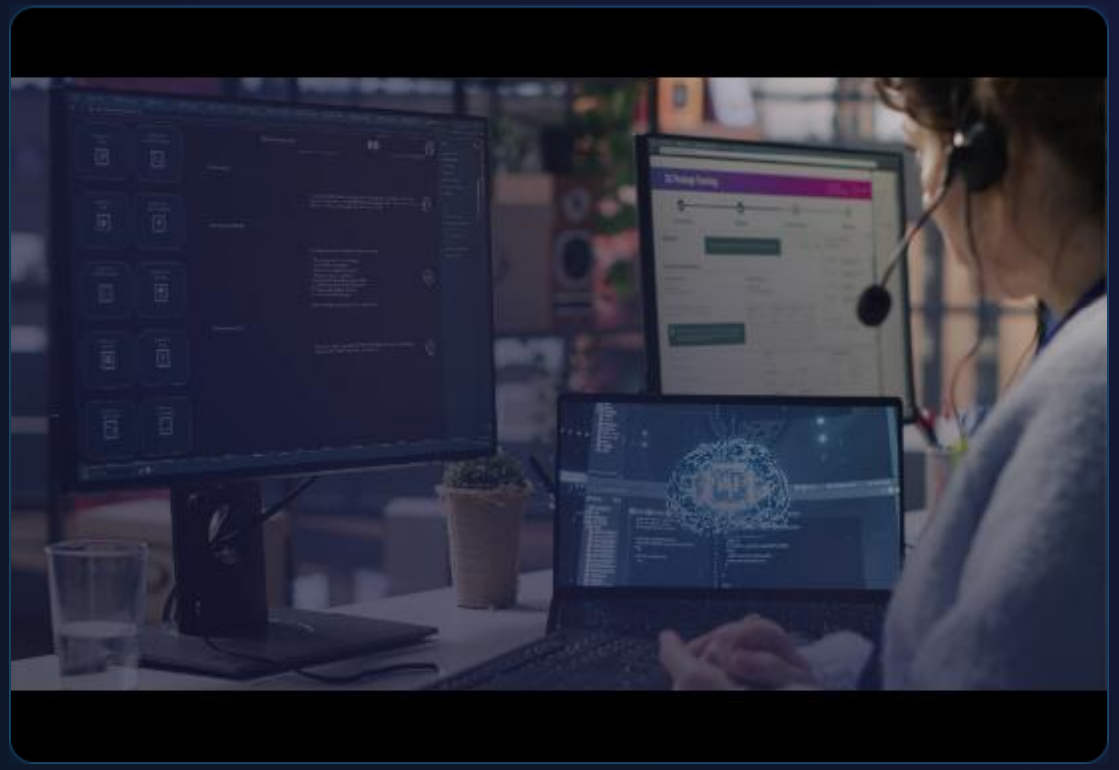
AI-Powered Test Strategy Generation

Automating QA documentation with Qwen3 and MCP

Github link:
yldavid2002/Test-Strategy-Advisor: Generate Test Strategy based on project introduction

The QA Challenge

- Modern software projects often begin with massive scope, unclear test strategy.
- Manually translating the project requirements into actionable test strategies require test expertise which is not affordable to many organizations.
- Maintaining standardized testing principles against the industry best practice is a constant battle.
- Relying on static, outdated templates severely limits adaptability to new testing methodologies.



The Automated Workflow

1. Ingest DOCX

Upload any project introduction document in DOCX format. Our system handles diverse structures, seamlessly extracting the core project scope, features, and technical requirements.

2. AI Digest

The system digests the complex context and intelligently outlines a comprehensive test strategy, strictly adhering to your organization's injected testing principles.

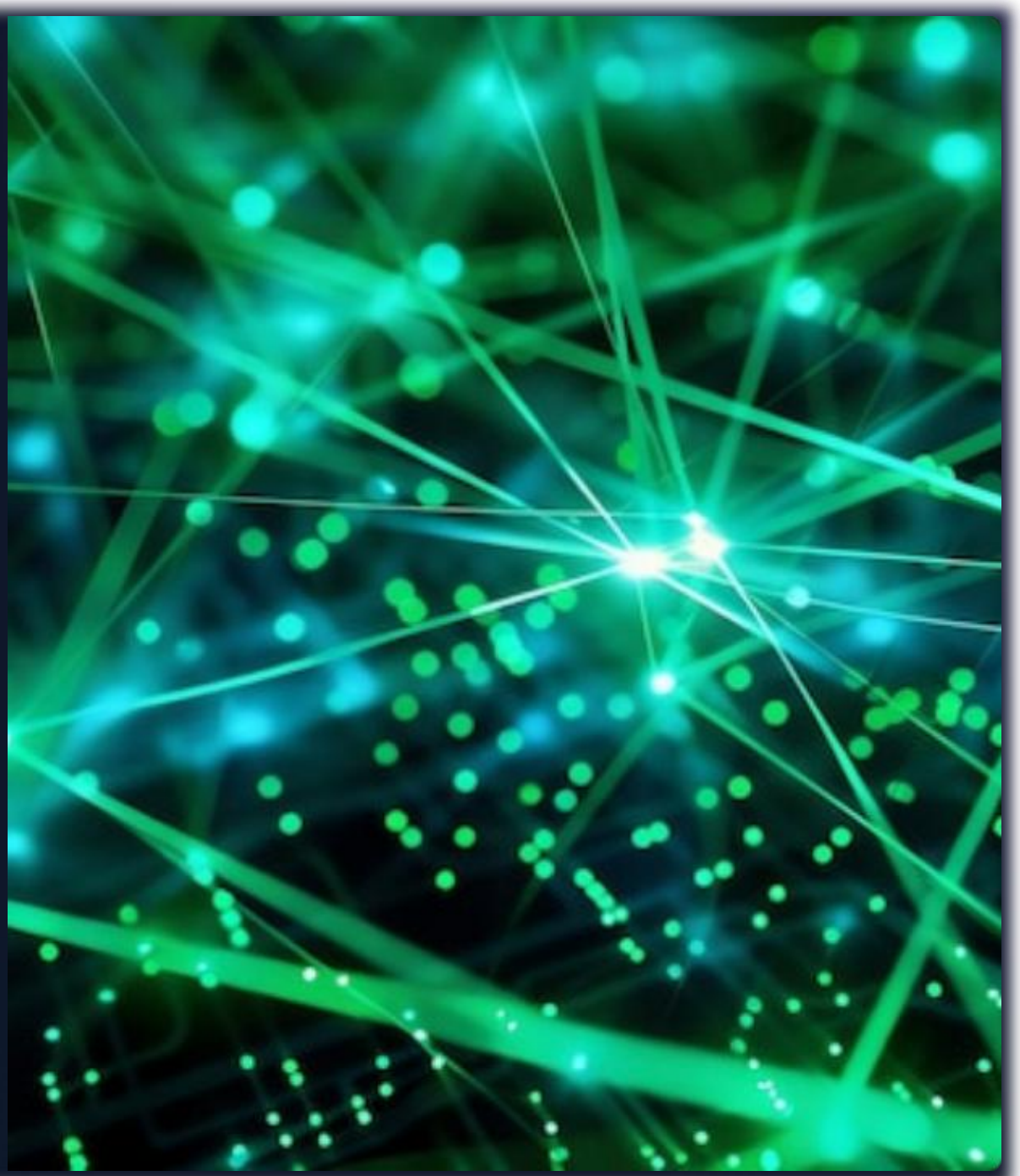
3. Export WORD

Instantly generate and export a fully structured, ready-to-execute test strategy back into a standard WORD file, saving your team hours of manual documentation.

Powered by Qwen3 & MCP

Our solution utilizes the cutting-edge **Qwen3 AI LLM** to intuitively parse and understand incredibly complex project requirements in natural language.

Crucially, we leverage an **MCP (Model Context Protocol) service** to strictly enforce core testing principles. This guarantees that the AI's output is not just creative, but procedurally accurate and fully compliant with enterprise standards.



Expert-Driven Scalability

100%

Modular Principles

Maintained by Domain Experts

One of our framework's most powerful features is architectural separation. The underlying AI engine is entirely decoupled from the specific testing logic. This means your **testing principles can be continuously updated and maintained by different testing experts over time**, ensuring the AI's output always reflects the absolute latest industry best practices.

Key Framework Benefits

Dynamic Intelligence

Unlike rigid automation scripts, Qwen3 adapts to linguistic variations in your DOCX inputs. The MCP service acts as a guardrail, combining the flexibility of LLMs with the strict compliance required by modern QA teams.

Compound Knowledge

As your QA organization grows, new domain experts can seamlessly inject specialized principles (like Security, Performance, or Accessibility) into the framework without altering the core AI pipeline.

Collaborative Maintenance

Democratized QA Logic

By relying on an MCP service for context injection, the responsibility of maintaining test quality is distributed. Security testers can update security principles, while UI/UX testers maintain usability rules. The AI dynamically combines these varying rulesets into every generated WORD document, acting as an intelligent force multiplier.

Ready to Automate?

Transform your project documents into comprehensive test strategies today.

QWEN3 + MCP TEST GENERATION FRAMEWORK

Demo

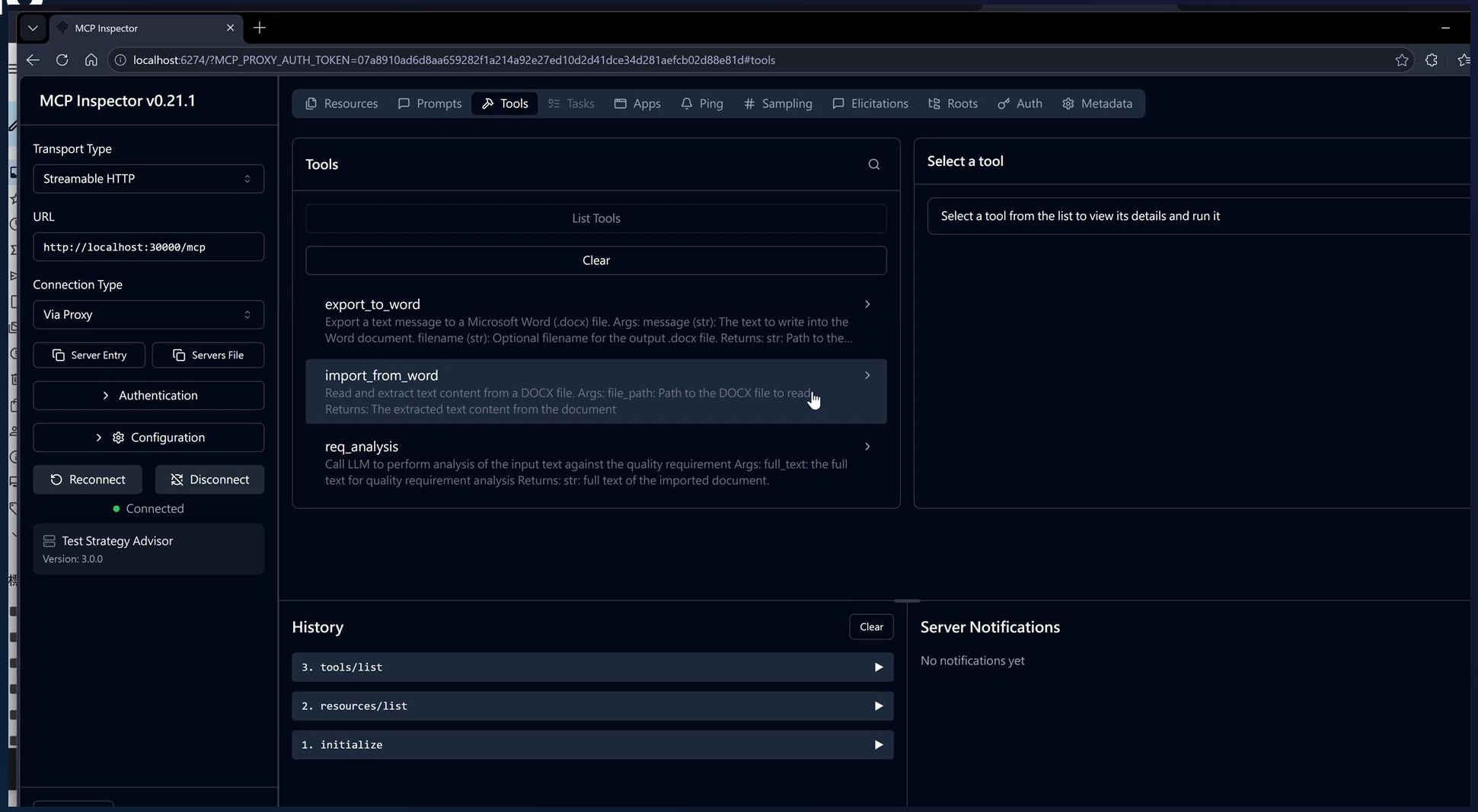
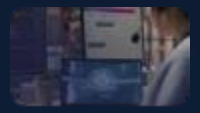

MCP Inspector
localhost:6274/?MCP_PROXY_AUTH_TOKEN=07a8910ad6d8aa659282f1a214a92e27ed10d2d41dce34d281aefcb02d88e81d#tools
MCP Inspector v0.21.1
Transport Type
Streamable HTTP
URL
http://localhost:30000/mcp
Connection Type
Via Proxy
Server Entry
Servers File
Authentication
Configuration
Reconnect
Disconnect
Connected
Test Strategy Advisor
Version: 3.0.0
Resources
Prompts
Tools
Tasks
Apps
Ping
Sampling
Elicitations
Roots
Auth
Metadata
Tools
List Tools
Clear
export_to_word
Export a text message to a Microsoft Word (.docx) file. Args: message (str): The text to write into the Word document. filename (str): Optional filename for the output .docx file. Returns: str: Path to the...
import_from_word
Read and extract text content from a DOCX file. Args: file_path: Path to the DOCX file to read Returns: The extracted text content from the document
req_analysis
Call LLM to perform analysis of the input text against the quality requirement Args: full_text: the full text for quality requirement analysis Returns: str: full text of the imported document.
Select a tool
Select a tool from the list to view its details and run it
History
Clear
3. tools/list
2. resources/list
1. initialize
Server Notifications
No notifications yet

Image Sources



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Source: www.qapitol.com

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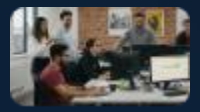
Source: www.freepik.com



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Source: overturepartners.com